PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila











PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Bacillus amyloliquefaciens CECT 5940: Unlocking Productivity Through Microbiome Modulation and Gut Health in Poultry

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Evonik Methionine (SEA) Pte., Ltd
Singapore

Bacillus amyloliquefaciens CECT 5940: Unlocking Productivity Through Microbiome Modulation and Gut Health in Poultry

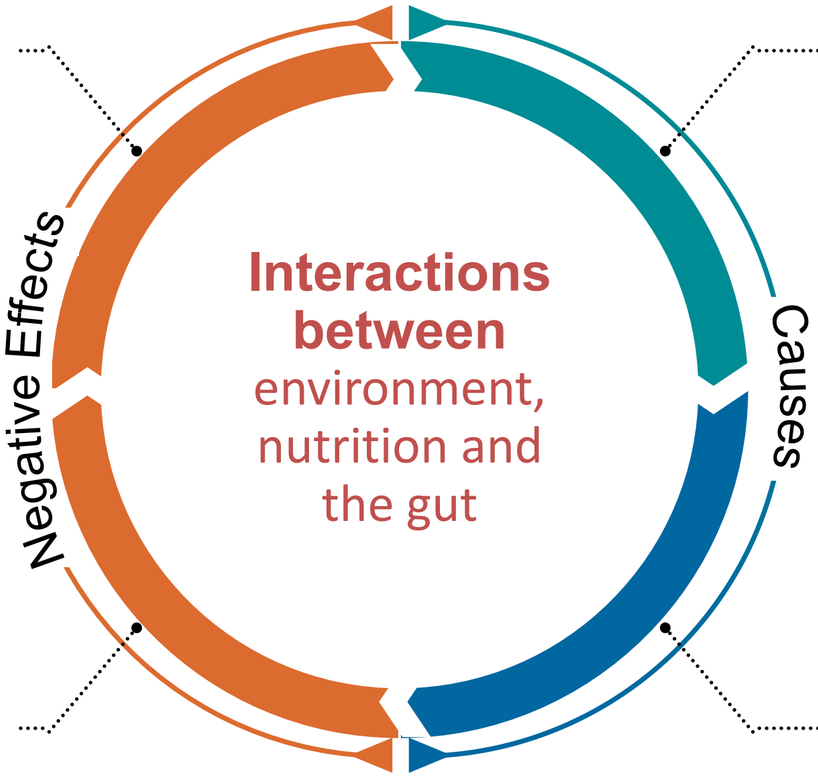
Dr. Balachandar Jayaraman BVSc & AH., PhD,
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Singapore

The key challenges in poultry production

- Growth performance ↓
- Flock uniformity ↓
- Mortality ↑
- Litter and air quality ↓
- Food pad lesions ↑
- Pathogen load in the GIT ↑
- Intestial lesions score ↑

Poor digestion & nutrient absorption

Morphological and functional alteration of the GIT



Feed & environment

Microbiota

- Heat stress
- Unbalanced diet
- Antinutritional factors
- Feed contaminants
- Coccidiosis
- Antimicrobial substance
- **Pathogenic bacteria**

Global challenge in poultry production causing significant economic damage

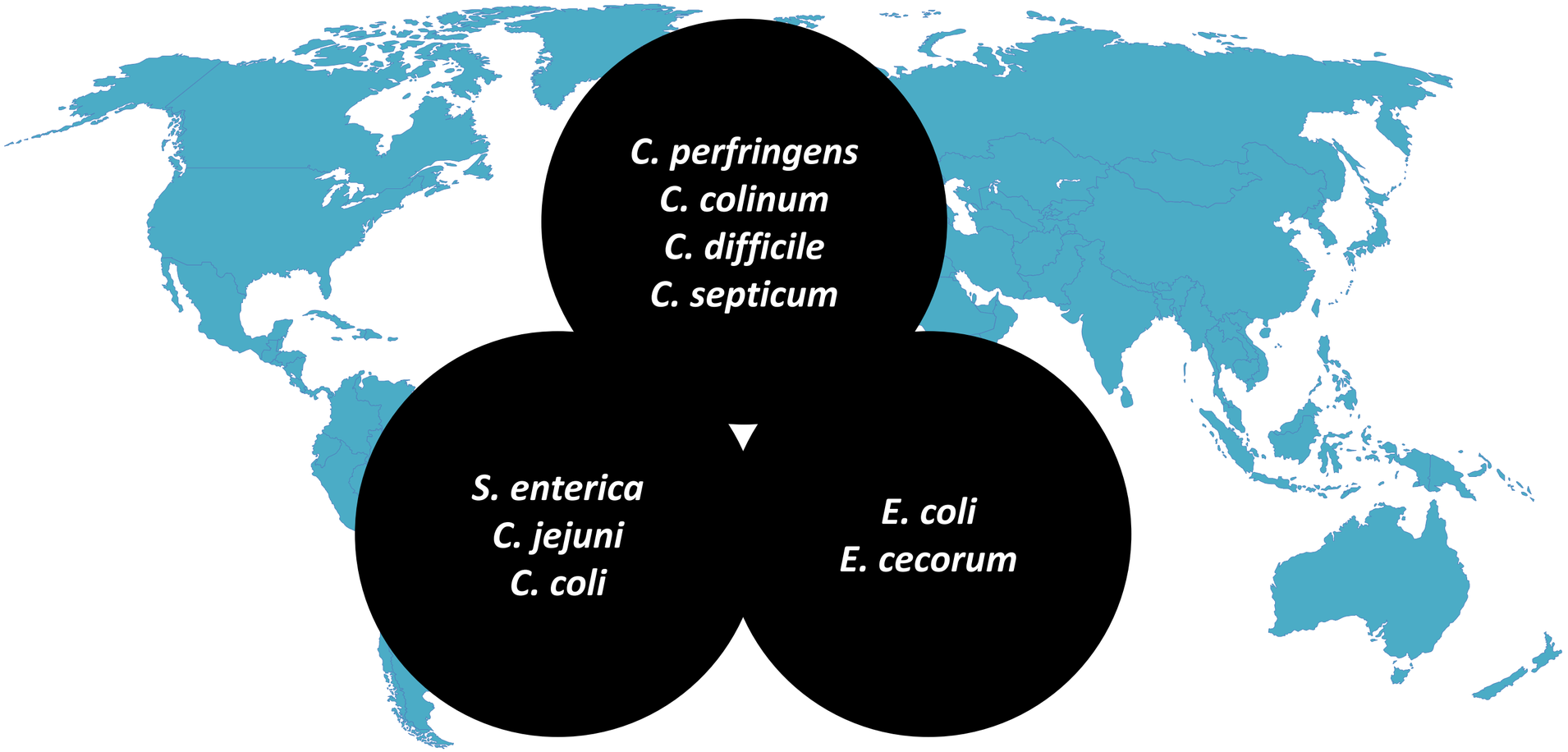
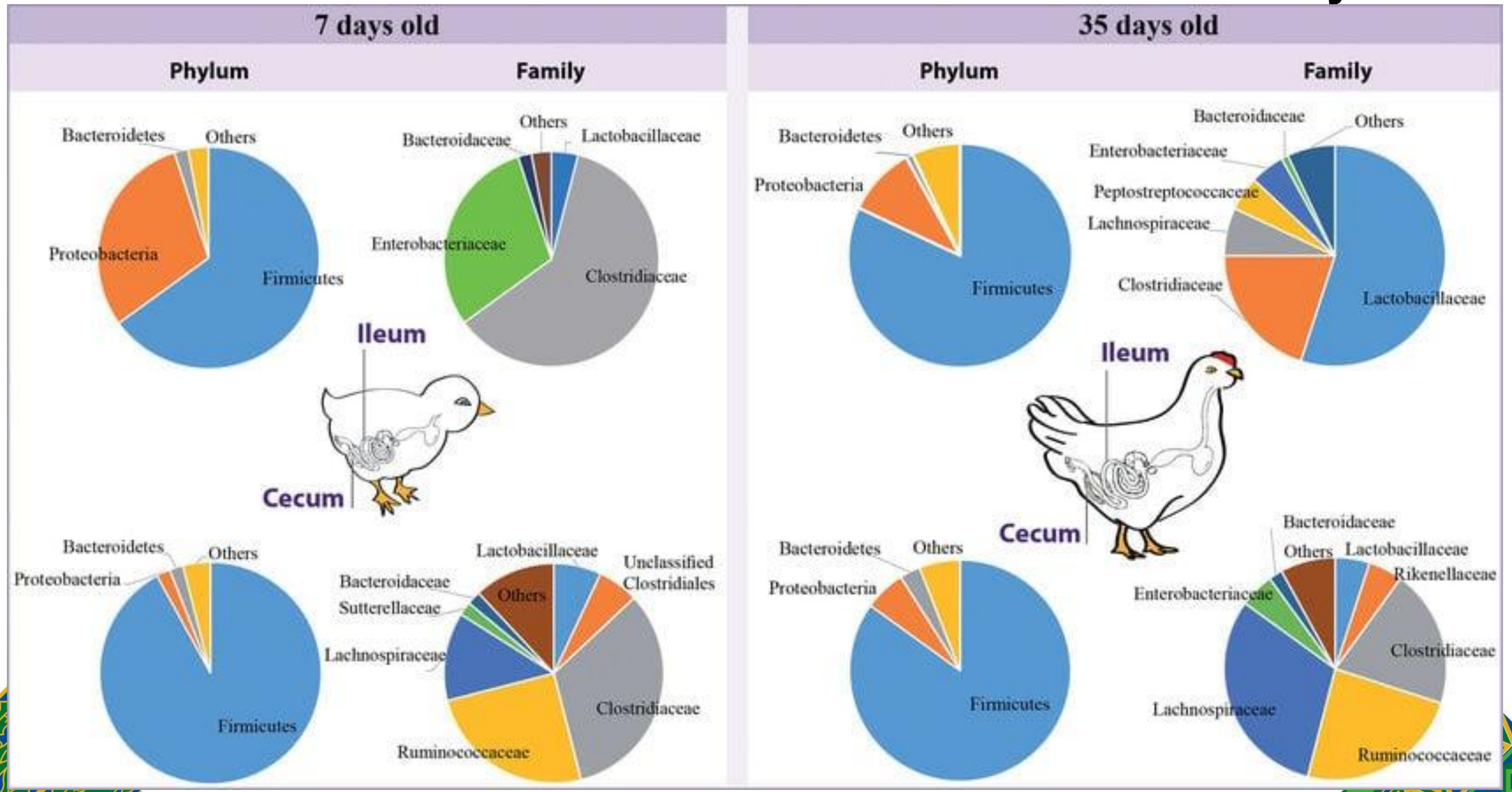


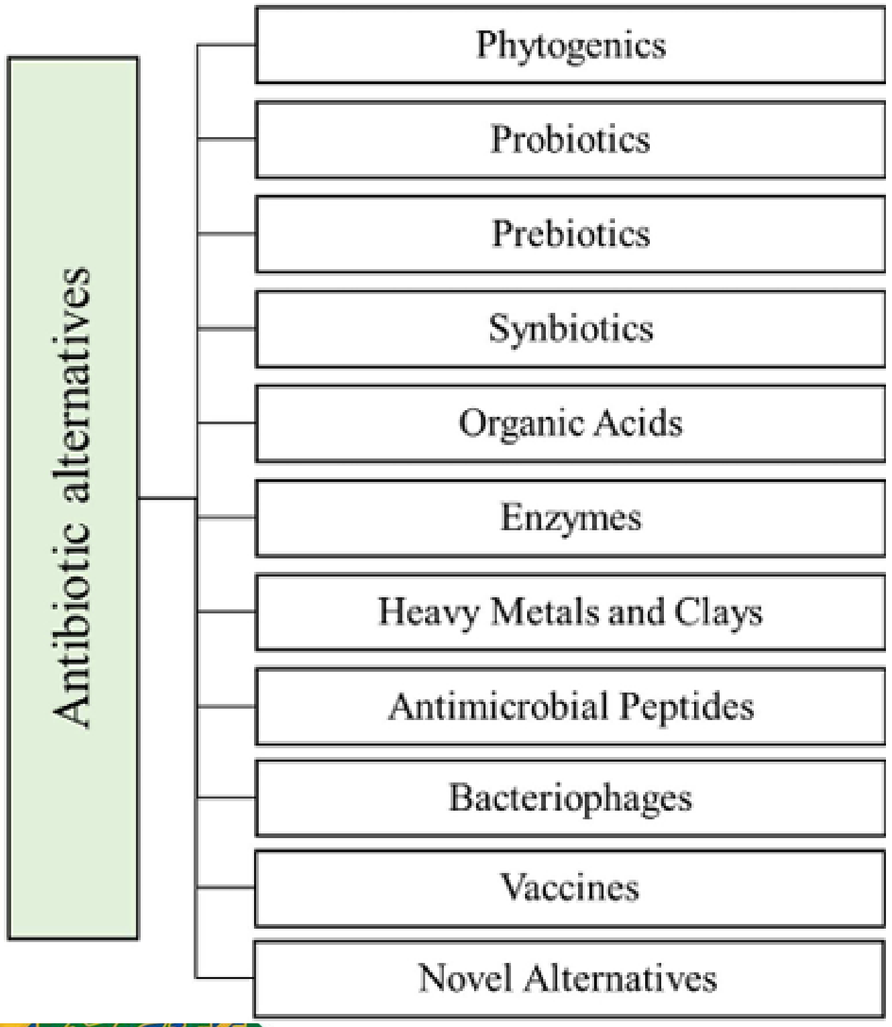
Figure: Global impact assessment of Clostridial enteritis, (Van de Sluis 2000)
https://www.efsa.europa.eu/en/topics/topic/salmonella
De Herdt et al., 2008 (Vlaams Diergeneeskundig Tijdschrift, 78)

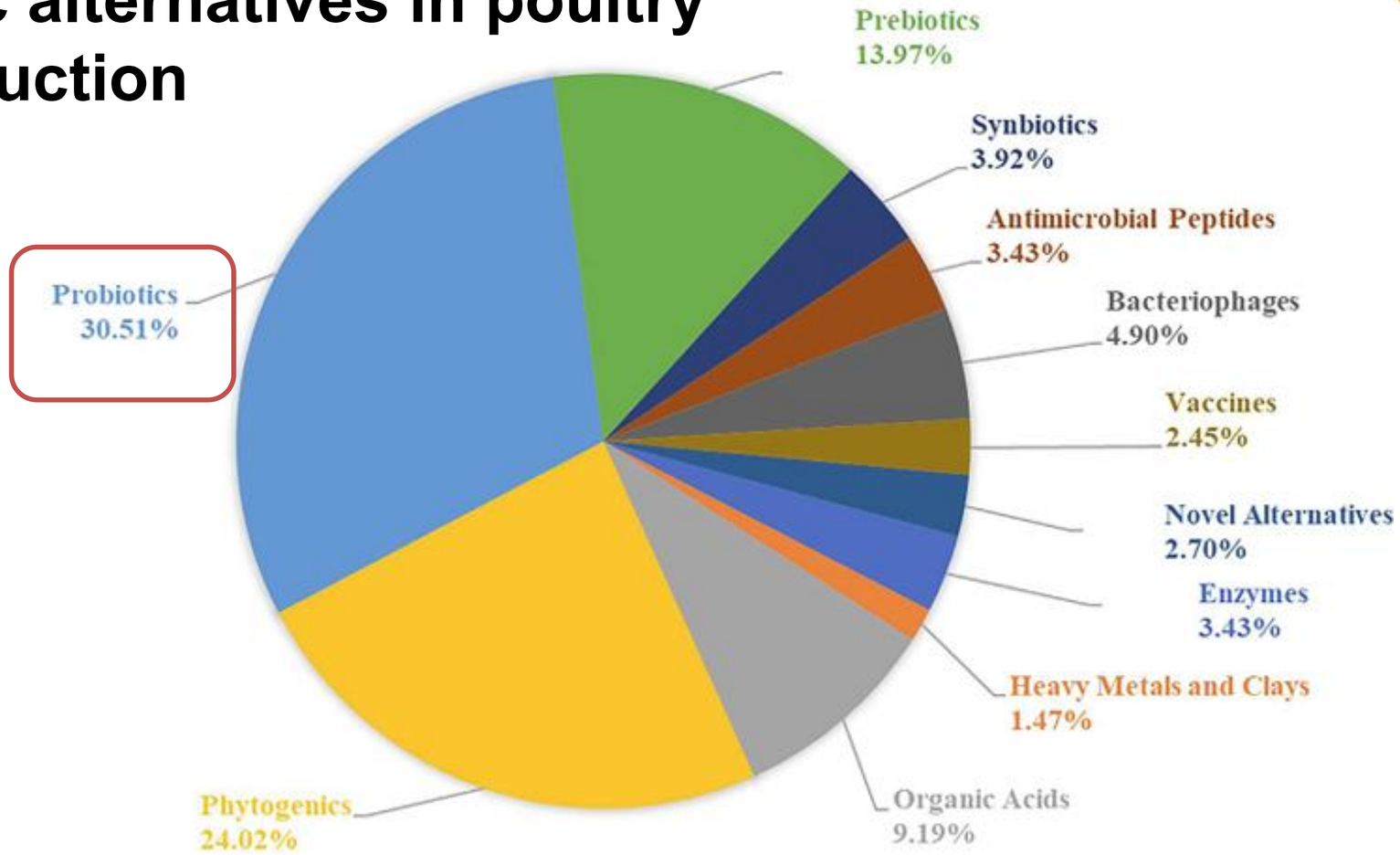
From Gut to Gain: The Microbiome's Contribution to Broiler Health and Productivity



Abedin and Zhao, 2015

Overview of antibiotic alternatives in poultry production



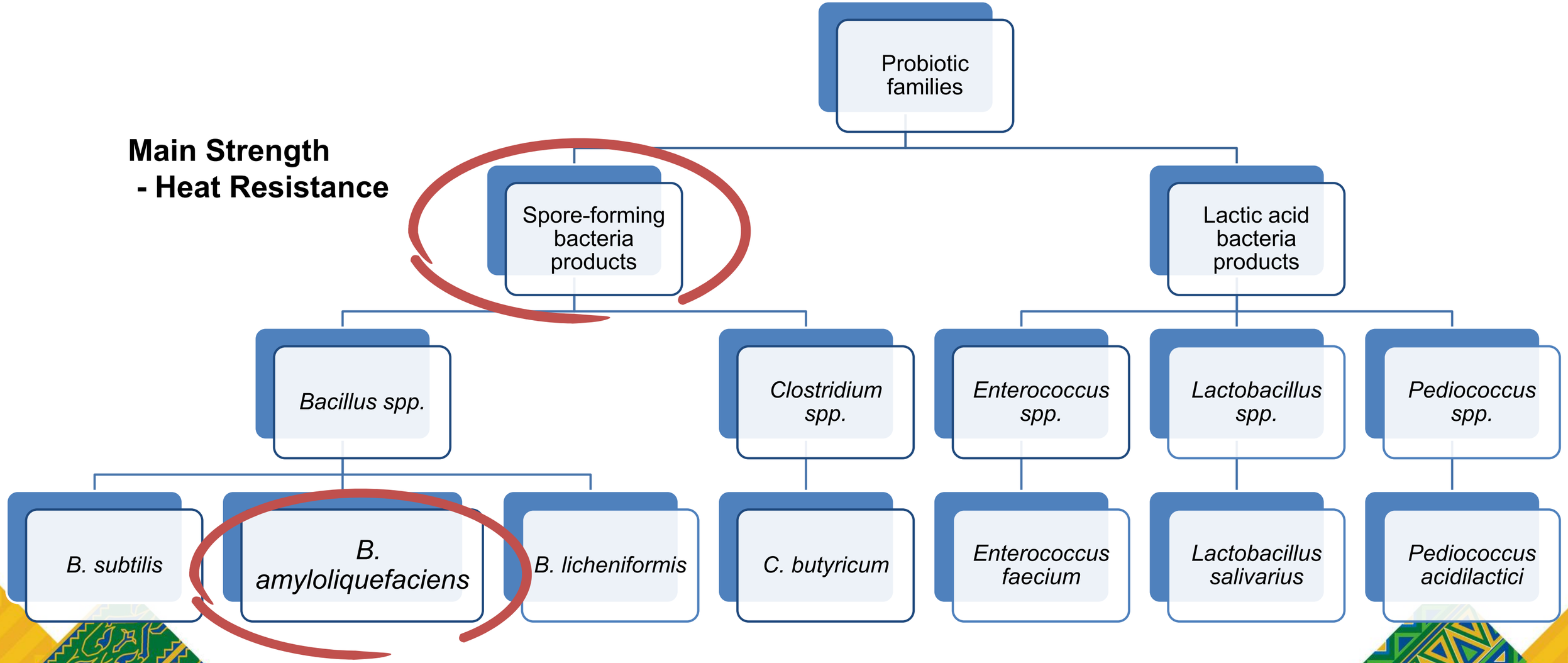


Proportion of publications that categorize alternatives to AGP in poultry

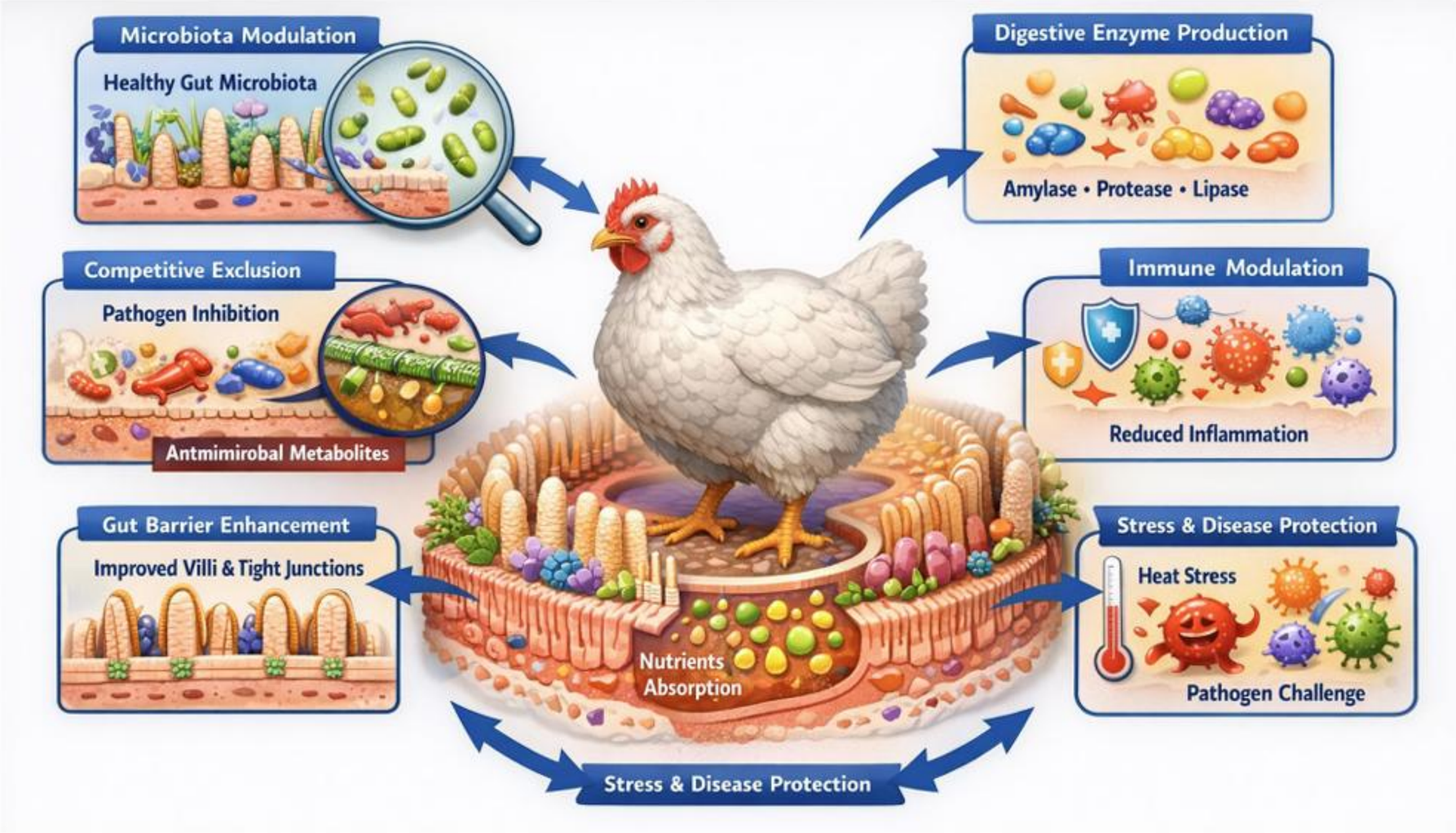
Wickramasuriya et al., 2024. Poultry Science Journal

Classification of Probiotics

Main Strength
- Heat Resistance



Winning the battle of intestinal peace with *Bacillus*—probiotics



Naeem and Bourassa, 2026

A multi-parameter strain screening was applied which resulted in the validation of *B. amyloliquefaciens CECT 5940 (BA 5940)*

Strain and spore characteristics

- Spore forming
- Heat stability 90°C
- Unique *Bacillus* strains:
  - *B. subtilis*
  - *B. amyloliquefaciens*
  - *B. licheniformis*
  - *B. pumilus*
  - *B. velezensis*



Growth

Strain fitness

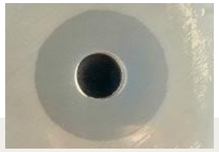
- Gizzard passage
- Bile resistance
- Osmotic stress
- Anaerobic growth
- Compatibility to organic acids
- Heat stability 95°C



Bile

Potential, safety & production

- Pathogen inhibition
- Enzyme production
- Metabolite expression
- AB susceptibility
- Strain robustness
- Genome analysis
- Production properties
- Benchmarking



Inhibition



***Bacillus amyloliquefaciens* CECT 5940**

Mode of actions of *B. amyloliquefaciens* CECT 5940

Pathogen inhibition :
C. perfringens, S. enteritidis, S. typhimurium, E. coli, C. difficile

Antimicrobial compounds :
Surfactin C, Macrolactin, Bacillaene, Bacilysin

Enzymes & Lactic acid production :
Xylanase, Cellulase, Oxidative stress protection enzymes

Quorum Quenching :
Block the communication of pathogens by AHL inactivation

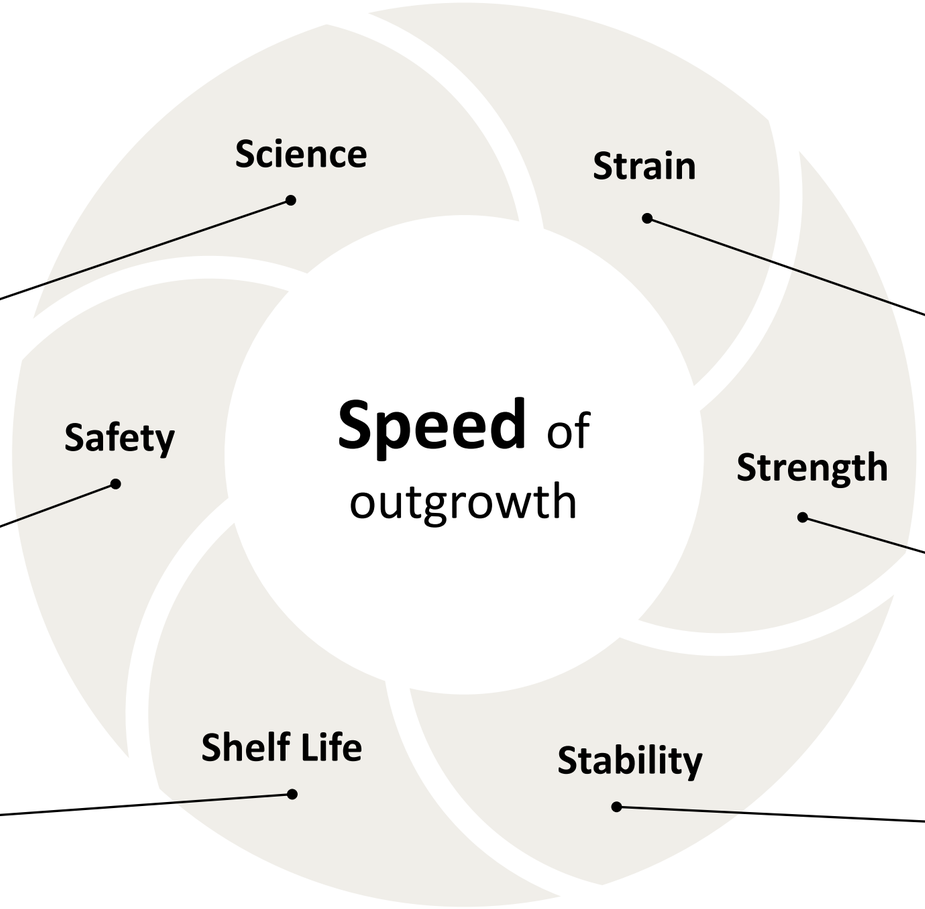
Microbiome shift :
Increases beneficial bacteria and decreases coliforms and *Clostridia* in feces

Immunomodulation :
Specific IL induction under challenged conditions

The challenge

Which probiotic is the right choice?

Spore-forming probiotics are not all the same



Science

Backed by robust research and validated by performance data.

Strain

Distinguished by strain specificity, as not all strains deliver the same benefits.

Safety

Defined by a proven safety profile, which is essential for both animal and human health.

Strength

Recognized for strong effects on animal welfare and performance.

Shelf Life

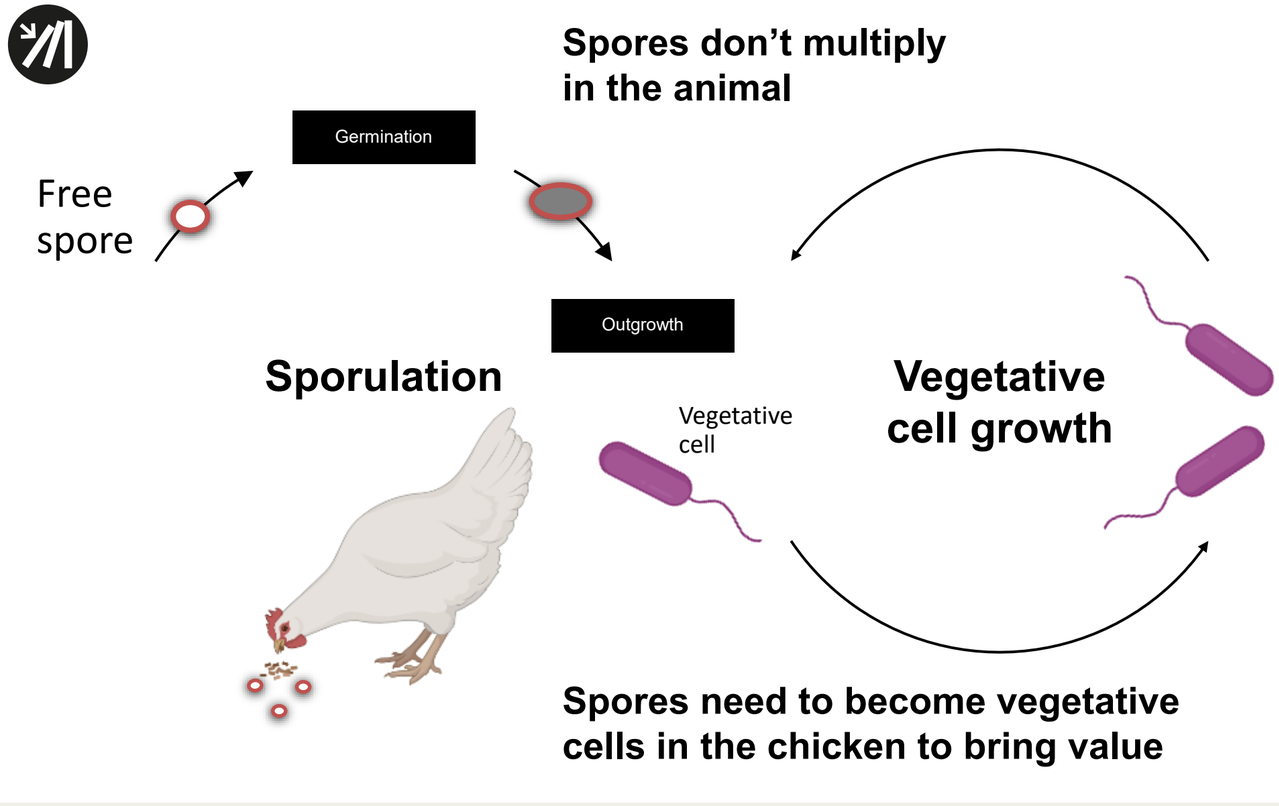
Characterized by long-term viability to ensure consistent performance.

Stability

Required to remain effective under processing and storage conditions.

The challenge

Reliable information



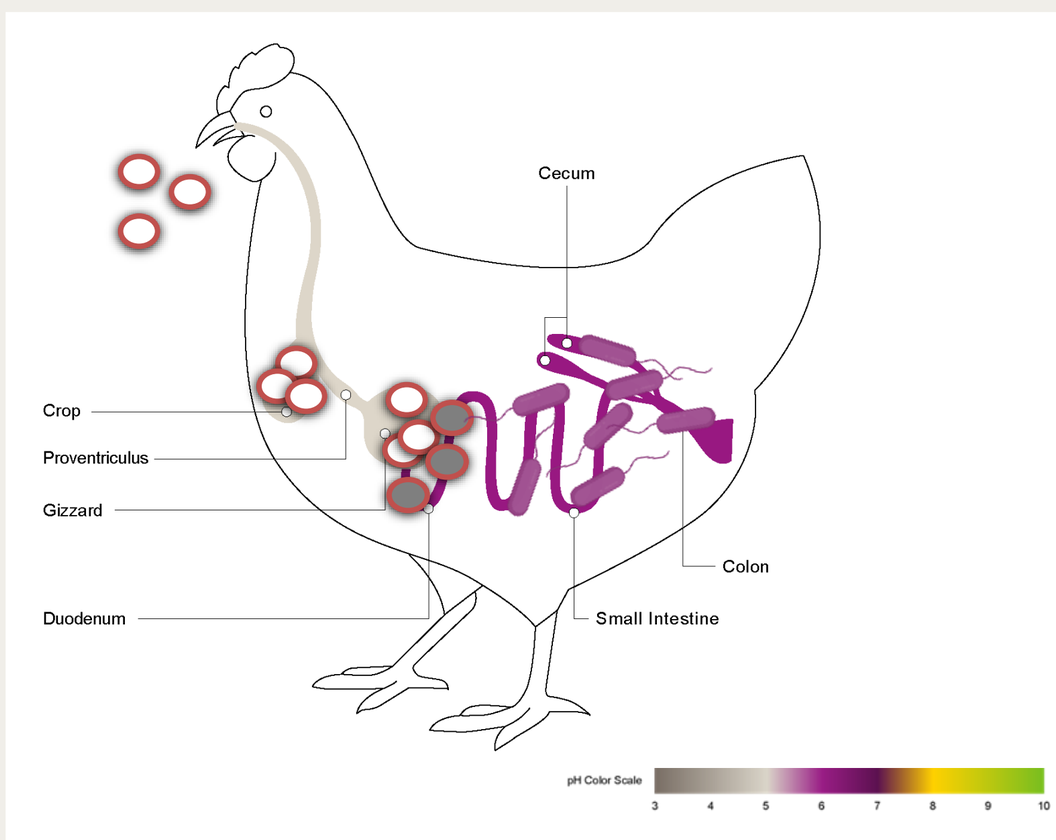
Spores

- Easy to store
- Stable to heat and pressure → survive pelleting
- Stable to challenging pH and trypsin → Survive stomach/ gizzard passage

Vegetative cells

- Inhibit the growth of **pathogens**
- Reduces the biofilm formation of ***E. coli***
- Production of **lactic acid** and **enzymes**
- Quorum Quenching: **blocks communication** among bacteria through quorum quenching

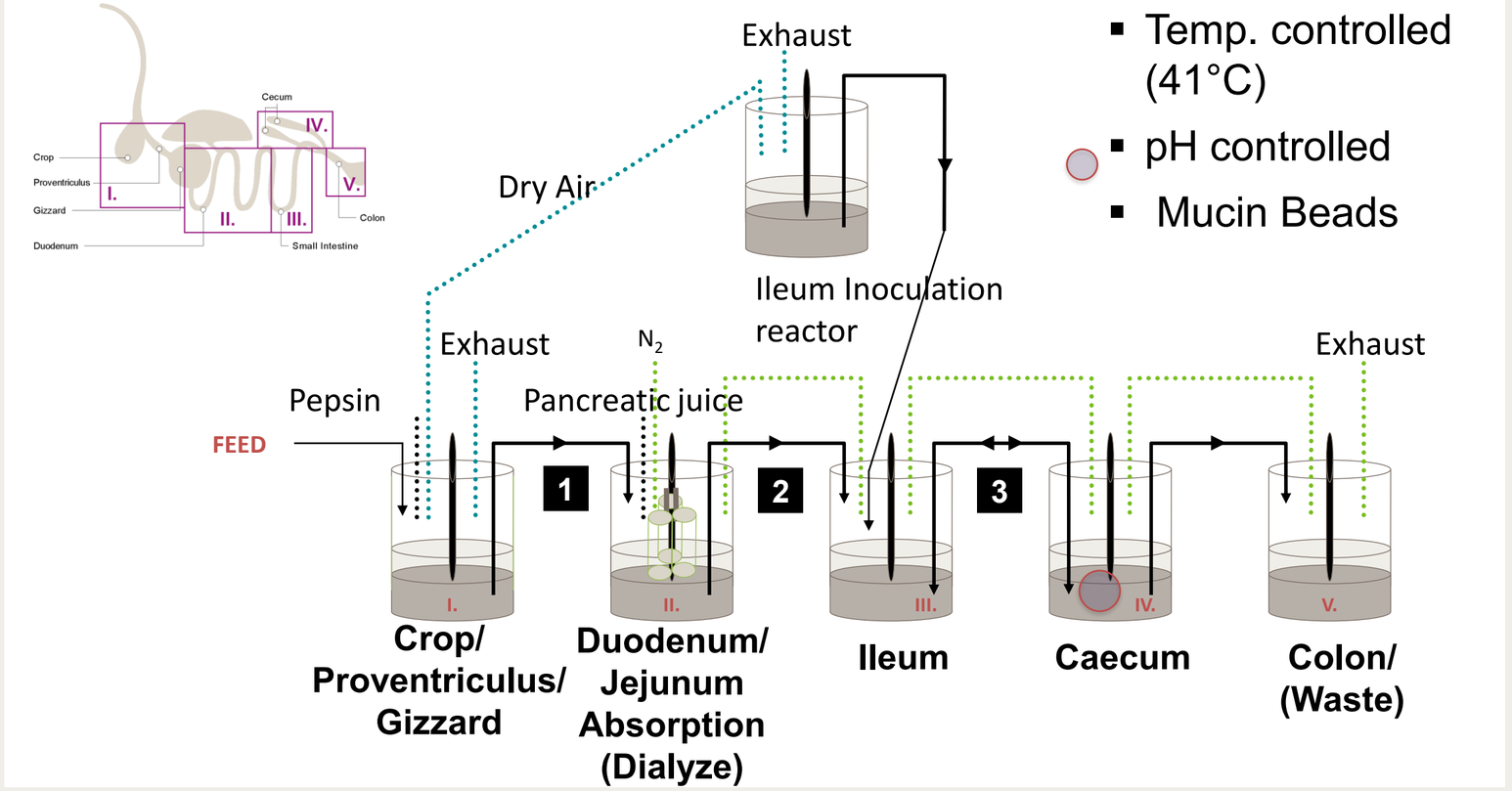
Germination and outgrowth of BA 5940 in chicken intestine



- Spores enter chicken GIT with feed
- Spores are stable against crop conditions
- Spores are stable during proventriculus/gizzard passage (low pH and pepsin)
- Spores are stable against pH up-shift, bile acid and pancreatin in duodenum
- **Release of amino acids & sugars from feed ingredients (Germinants) activate germination of BA 5940 spores in the intestine**
- **In the ileum BA 5940 is present as vegetative cells**

DAISy – Dynamic Avian Intestine *in-vitro* System

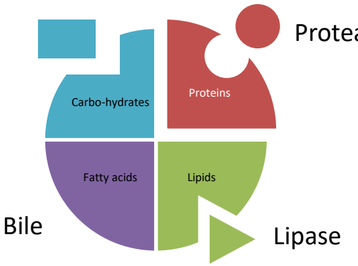
Test system that simulates all GIT challenges under proper conditions



- Temp. controlled (41°C)
- pH controlled
- Mucin Beads

1

Amylase

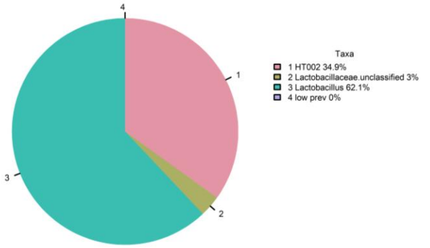


Protease

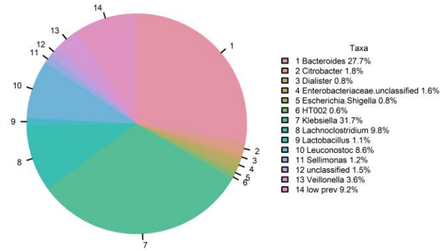
Bile

Lipase

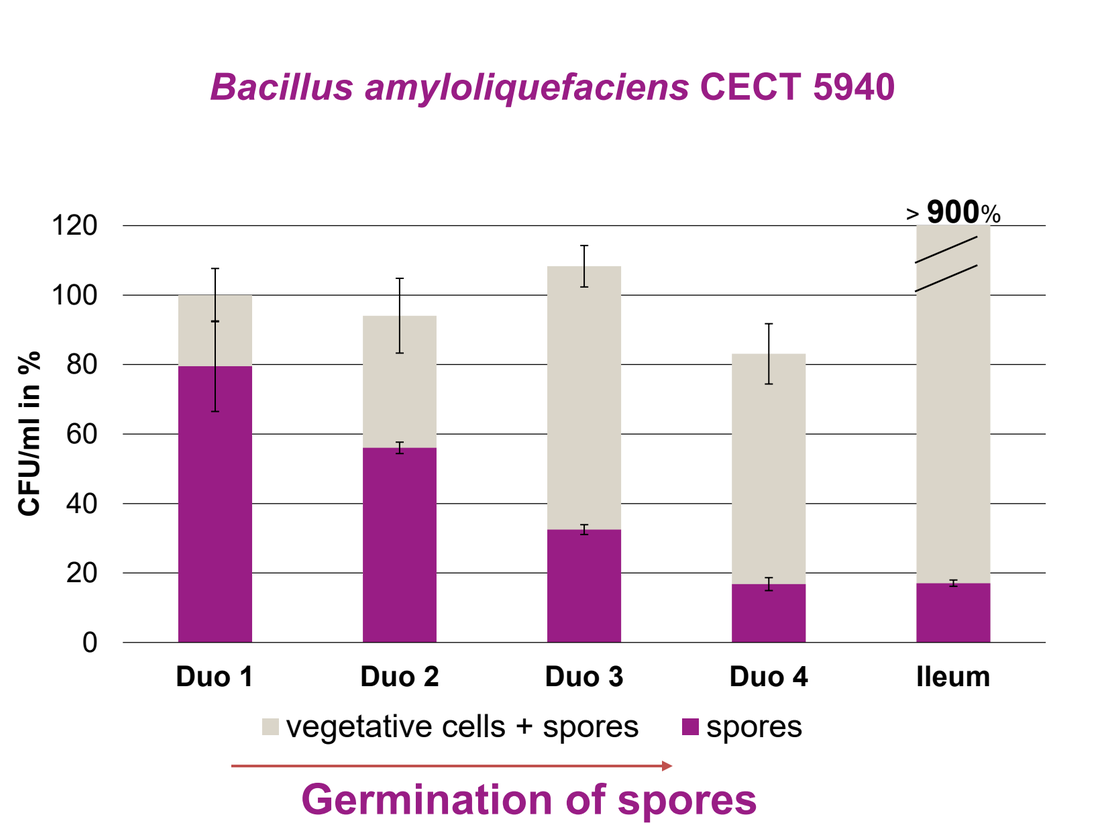
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3



Germination of spore product determined in DAISy

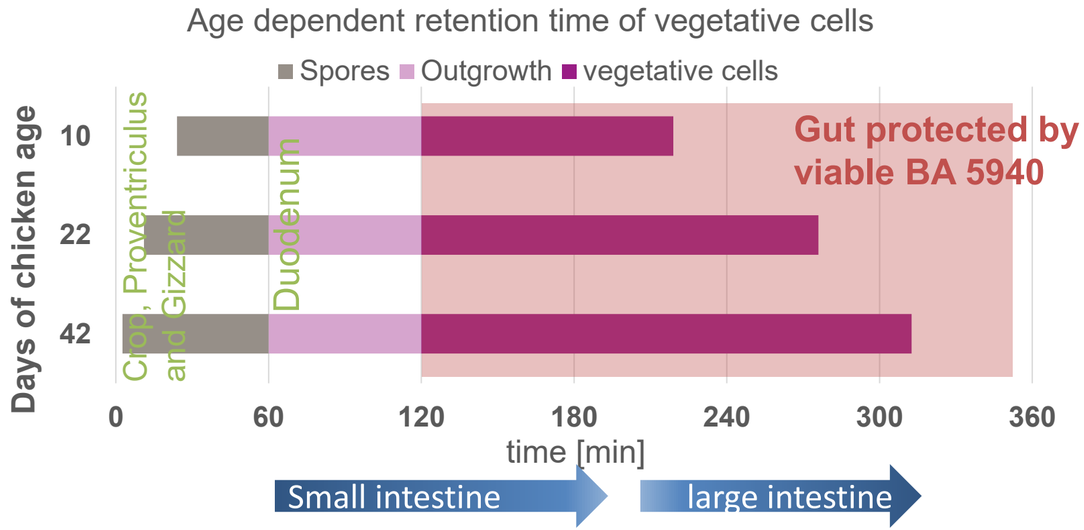


- Total aerobic microbes (spores + vegetative cells) number keeps stabile, but spore numbers are decreasing
  - Spores are germinating and become vegetative cells
- **BA 5940 spores germinate efficiently in the nutrient rich duodenum/jejunum segment (activation of germination)**
- **In the ileum BA 5940 is present as vegetative cells**
- Ba 5940 germination takes approx. 60 min under realistic simulated conditions in DAISy

The digestive cycle of chicken is very fast and depends on age



Age	10 days old	22 days old	30 days old	42 days old
Daily feed intake	40 g	93 g	117 g	160 g
Broiler weight	~ 400 g	~ 850 g	~ 1500 g	~ 2800 g
Retention time	**3h 15 min**	**4h 25 min**	**4h 44 min**	**5h 10 min**

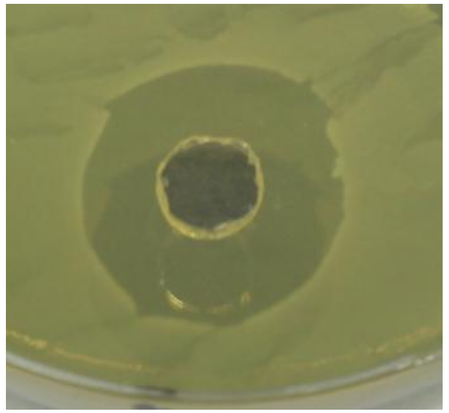


Spore outgrowth time is crucial, as it determines the duration of effectiveness during gastrointestinal transit

***Bacillus amyloliquefaciens* CECT 5940**
Pathogen inhibition



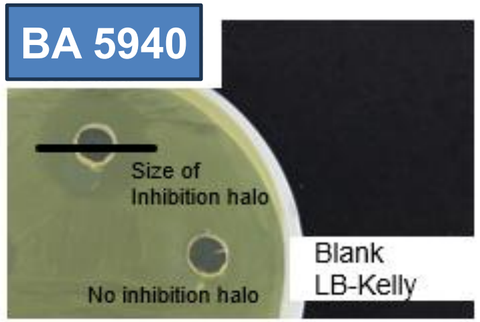
Probiotic BA 5940 inhibits common pathogenic organisms in poultry



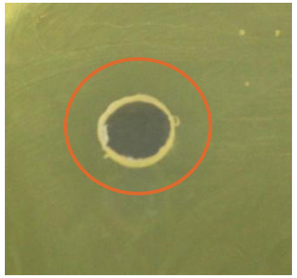
C. perfringens ATCC 13124

BA 5940 can inhibit

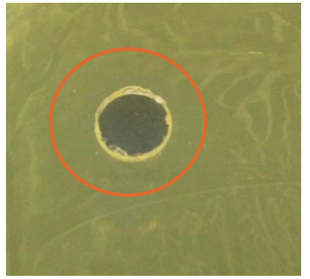
C. perfringens type strain ATCC 13124 and several field isolates very effective



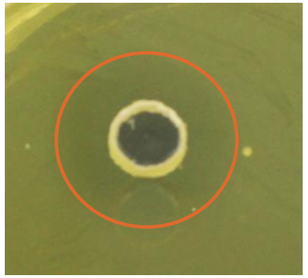
BA 5940 can inhibit all tested ***E. coli*** strains very effective



S. gallinarum DSM13674



S. gallinarum ATCC9184



S. pullorum

BA 5940 can inhibit both ***Salmonella*** **from Typhoid group**

Brazil Broiler– Probiotic BA 5940

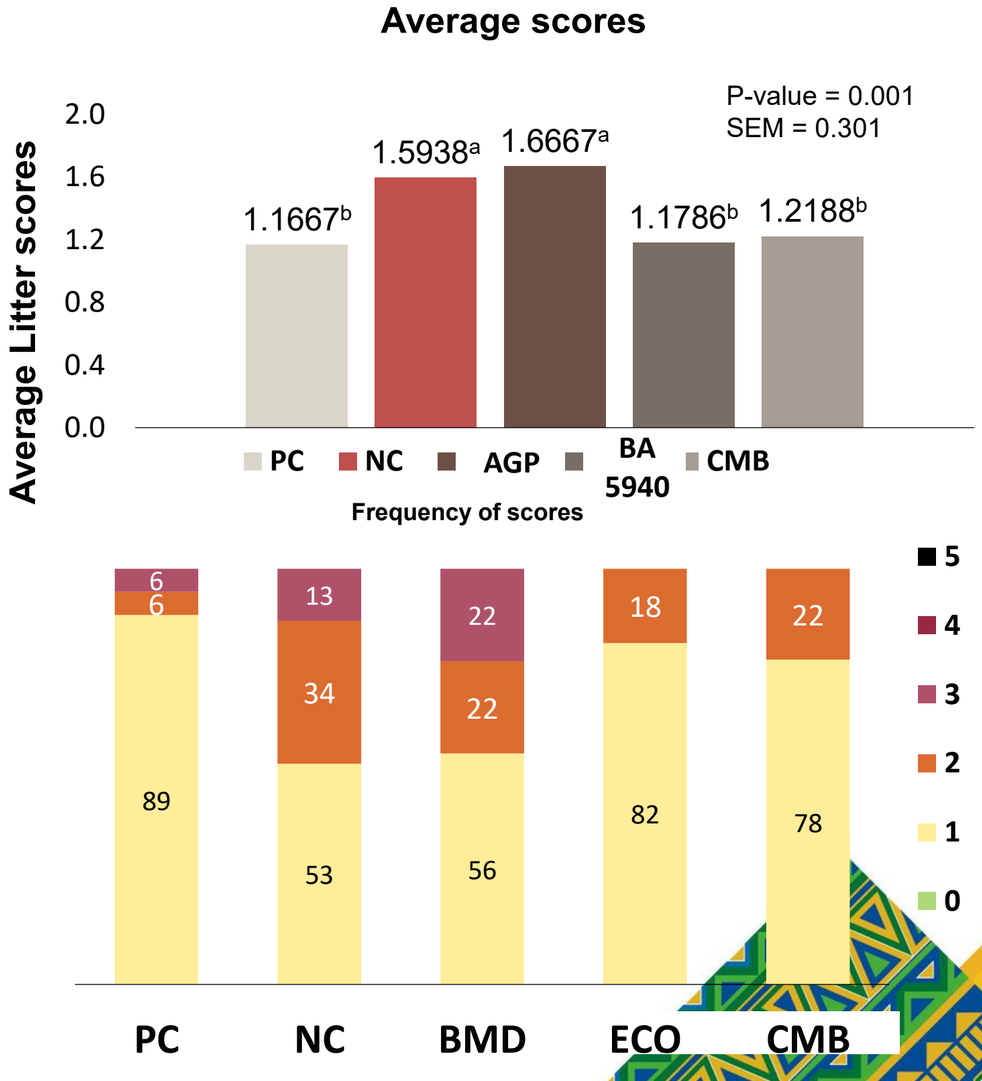
C. Perfringens inhibition

Treatment	Ileum (log CFU/g)		Cecum (log CFU/g)	
	21 days	28 days	21 days	28 days
Non-challenged	5.36[b]	5.25[b]	5.84	5.71
Challenged	5.85[a]	5.70[a]	6.11	5.92
Challenged + AGP	5.07[b]	4.82[b]	5.49	5.35
Challenged + BA 5940	5.24[b]	4.99[b]	5.68	5.43
Challenged + BA 5940 + AGP	5.17[b]	4.90[b]	5.53	5.40
SEM	0.420	0.355	0.481	0.487
P value*	0.003	<0.0001	0.060	0.076

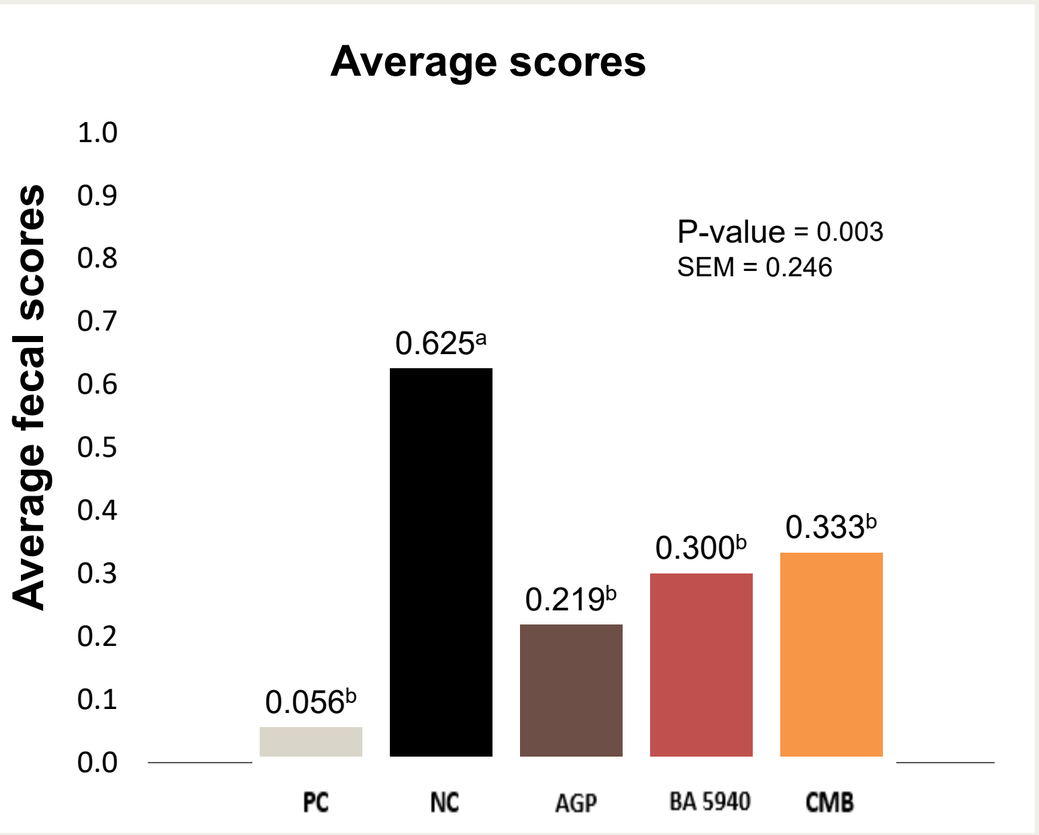
Data was analyzed by one-way ANOVA and all statistical presupposition were evaluated followed by multiple comparisons of means using the scheffe test and a p-value < 0.05 was deemed significant (SAS v9.4, SAS Institute).

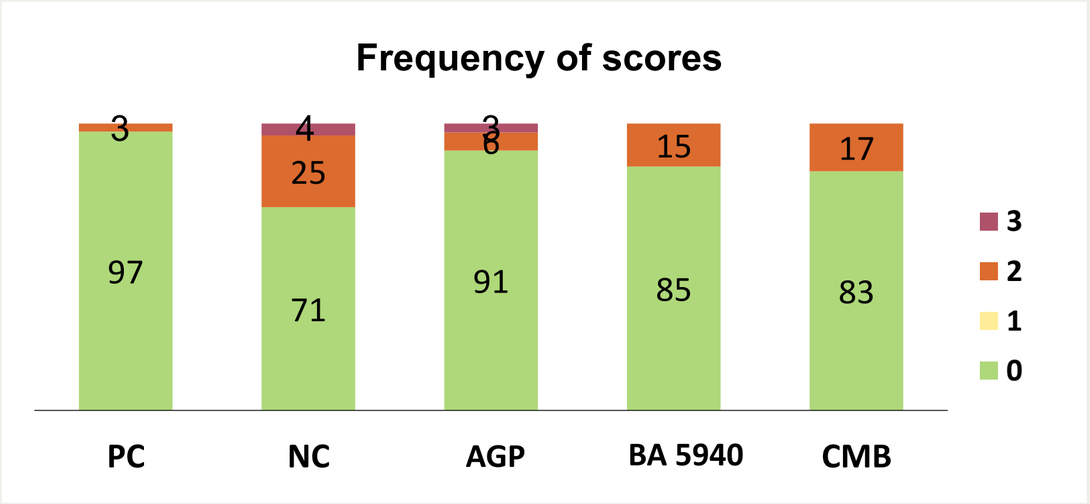
Supplementation with probiotic BA 5940 reduces significantly the amount of pathogenic strains of *C. perfringens* in both ileum and cecum.

BA 5940 significantly improved the litter quality



BA 5940 significantly improved fecal consistency





Fecal scores


Score 0


Score 2


Score 3

Spain Broiler Trial– BA 5940

Salmonella spp. inhibition

100 healthy Ross broiler chickens (male and female)

42 days

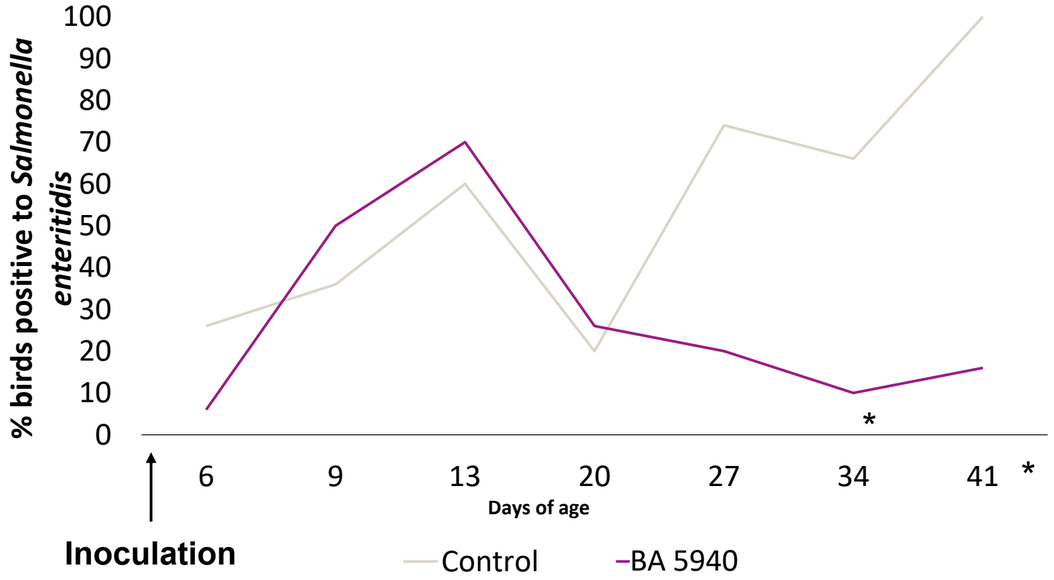
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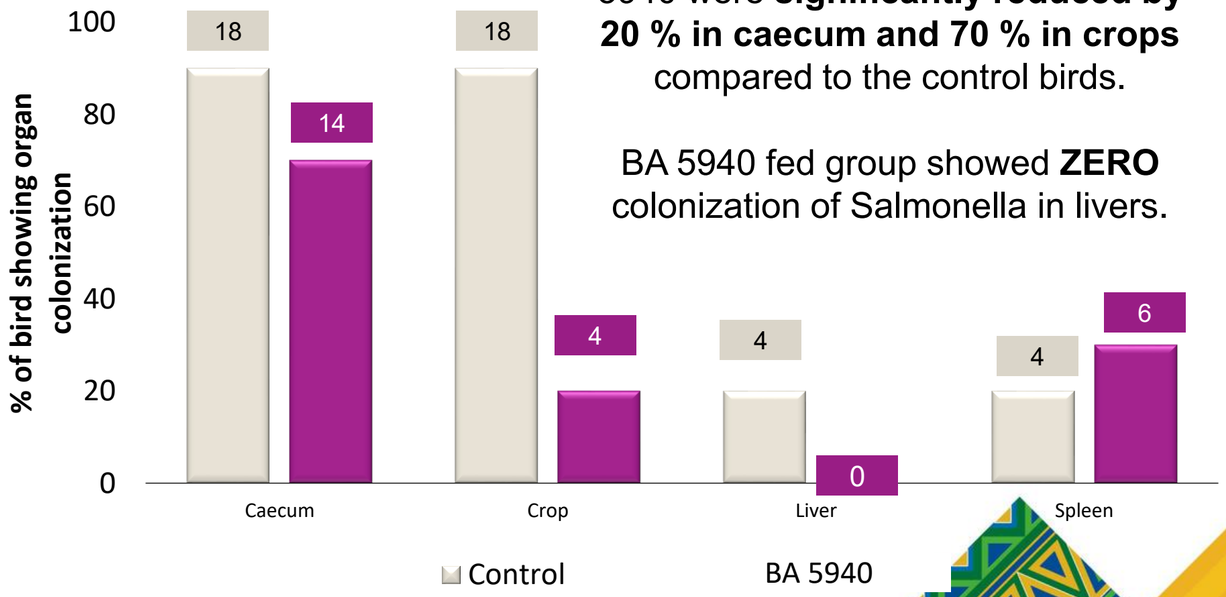
- Control basal diet
- Basal diet supplemented with BA 5940 (1x1E+9 CFU/g) at 1 kg/ton feed

On day 5, 20% of the birds per experimental group were orally inoculated with *Salmonella enterica* serovar *enteritidis* (1E+5 CFU/bird) and then placed back into the pens in order to simulate horizontal infection

- At days 6, 9, 13, 20, 27, 34 and 41 of the experiment (1, 4, 8, 15, 22, 29 and 36 days post-challenge), cloacal swabs were collected from all birds to measure *S. enteritidis* counts
- At day 42 all birds were slaughtered and crop, caecum, liver and spleen were collected for further bacteriological analyses

BA 5940 significantly reduced shedding of *Salmonella enteritidis*





Salmonella counts in the birds fed BA 5940 were **significantly reduced by 20 % in caecum and 70 % in crops** compared to the control birds.

BA 5940 fed group showed **ZERO** colonization of Salmonella in livers.

Meta-Analysis of BA 5940 Trials

	Study 1	Study 2	Study 3	Study 4	Study 5	Study 6	Study 7	Study 8	Study 9	Study 10	Study 11	Study 12
Location	Spain I	Spain II	Spain III	Thailand I	Thailand II	Romania	Czech Republic	Brazil	Australia I	Australia II	Australia III	USA
Broiler Strain	Ross 308	Ross 308	Cobb 500	A.A. Plus[1]	Cobb 500	Ross 308	Ross 308	Cobb 500	Ross 308	Ross 308	Ross 308	Cobb 500
Reps x Birds[2]	12 x 12	13 x 60	6 x 10	24 x 10	6 x 20,240	25 x 6	6 x 35	9 x 34	8 x 15	8 x 15	8 x 15	5 x 20
Duration	42 days	42 days	42 days	35 days	45 days	35 days	35 days	42 days	35 days	35 days	35 days	42 days
Diet base	Wheat, barley, SBM (mash)	Wheat, barley, Corn, SBM (mash)	Corn, SBM (pellet)	Corn, SBM (pellet)	Corn, SBM	Corn, wheat, SBM (Mash)	Corn, wheat, SBM (pellet)	Corn, SBM, MBM (mash)	Wheat, sorghum, SBM (pellet)	Wheat, sorghum, SBM (pellet)	Wheat, sorghum, SBM (pellet) Reduced CP[3]	Corn, SBM (mash/ pellet)
Conditions/ Challenge	Battery cages	Pens	Pens	Pens	House	Pens	Pens	Pens/ NE challenge[4]	Pens	Pens/NE challenge[5]	Pens/NE challenge[5]	Pens/ST challenge[6]
Medication	No	No	No	No	Tylosin + Amoxicillin[7]	No	No	No	No	No	No	Salinomycin sodium[8]
BA 5940 inclusion	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	50g/1000 L water daily	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed	$1x10^6$ CFU/g feed

Abbreviations: SBM (soybean meal); MBM (meat and bone meal)

[1] A.A. Plus, Arbor Acres Plus.

[2] Reps x Birds, number of replicates and number of birds per replicate

[3] Reduced CP (crude protein), diet had 2% less CP than diet from study 10. CP reduction was achieved by the supplementation of DL-Methionine, L-Lysine HCL, L-Threonine, L-Valine, L-Isoleucine, and L-Arginine.

[4] NE challenge, Necrotic Enteritis model: day 17: 50,000 oocysts E. maxima; days 18, 19, 20: 2.5 X 10^6 CFU of Clostridium perfringens.

[5] NE challenge, Necrotic Enteritis model: day 9: 5,000 oocysts of E. maxima, E. acervulina, 2,500 oocysts of E. brunetti; days 14 and 15: 10^8 CFU of Clostridium perfringens.

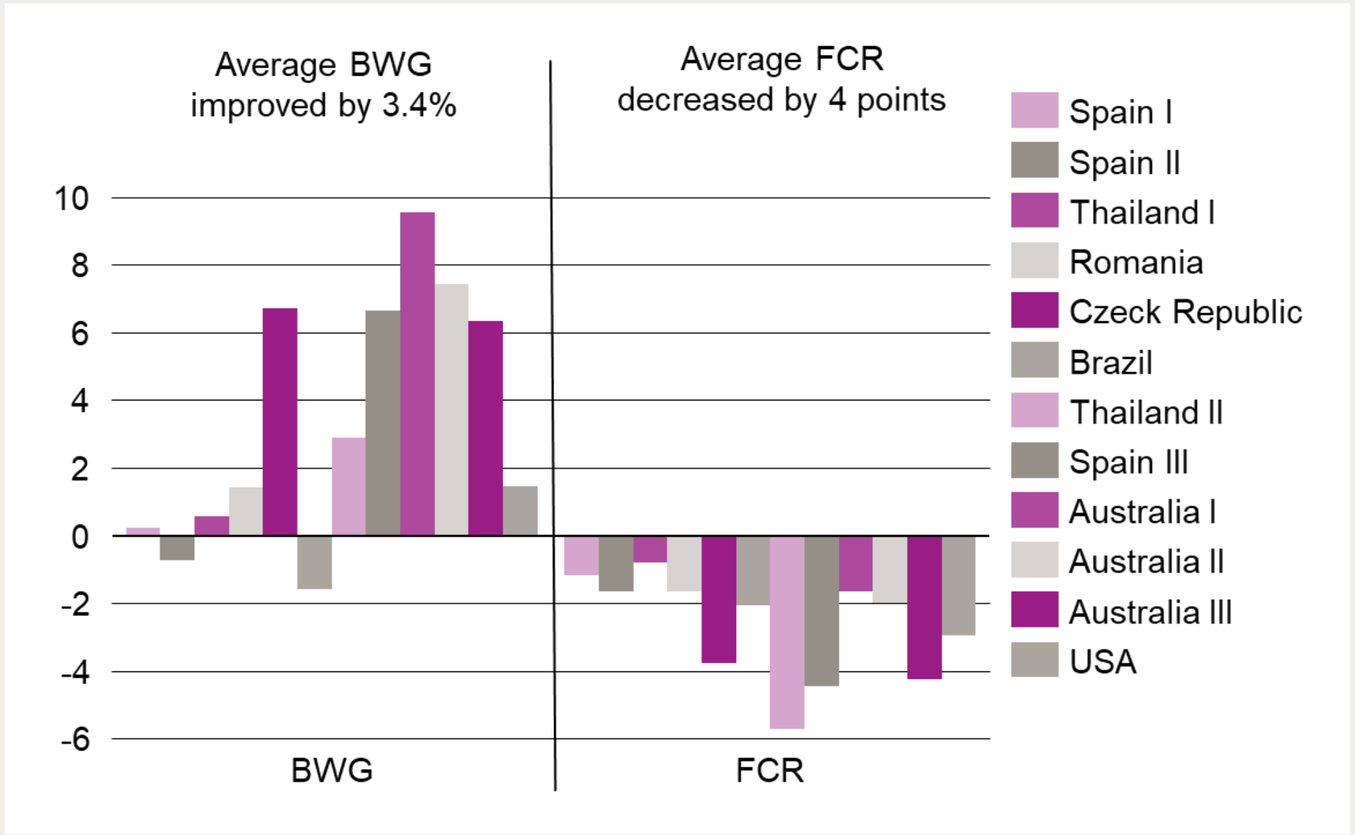
[6] ST challenge, *Salmonella* Typhimurium challenged birds (50%) were introduced into pens at day 7.

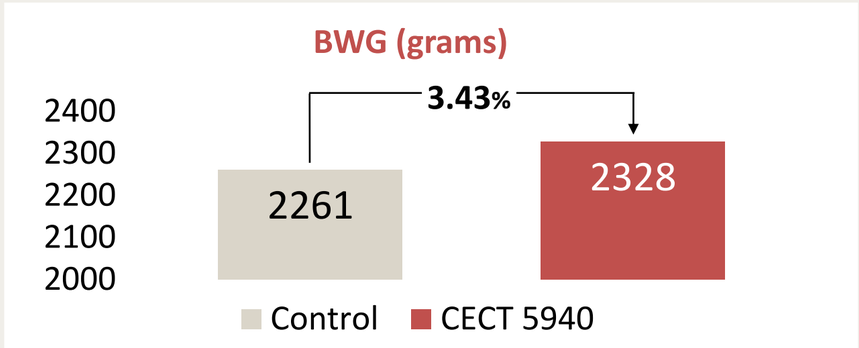
[7] Tylosin and Amoxicillin were applied in the drinking water from days 2 to 6 and days 9 to 11.

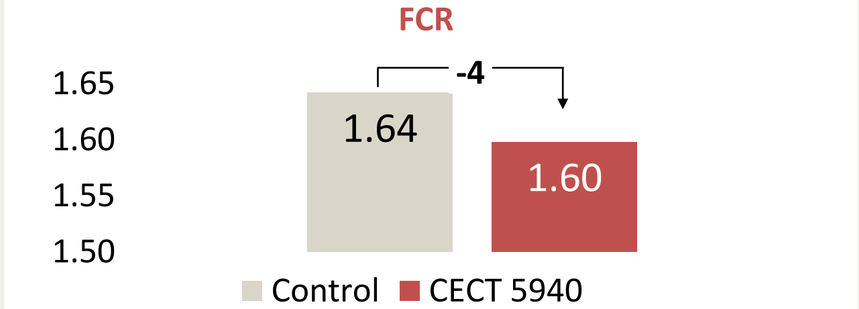
[8] Salinomycin sodium was mixed in feed for all phases.

Results

Overview of performance improvement







Probiotic BA 5940 commercial broiler trials in Thailand: Performance

Commercial Broilers/Thailand

May-Dec 2025

Objective : To evaluate the effects of BA 5940 on improving performance and gut microbiome modulation in commercial broilers

	Group	Breed	Sex	Bird/house	Age (day)	BA 5940 Program* (day)	Final BW (g)	ADG (g/d)	FCR	Livability (%)	Performance Index (PI)
Farm 1	Control	ROSS 308	Male	27,000	40	-	2,960	72.20	1.59	93.15	445
	BA 5940	**ROSS 308**	**Male**	**27,000**	**40**	**1-40**	**3,030**	**75.75**	**1.56**	**96.56**	**469**
Farm 2	Control	Arbor Acres	Male	24,072	41	-	3,026	73.81	1.59	92.18	428
	BA 5940	**Arbor Acres**	**Male**	**24,072**	**41**	**19-41**	**3,060**	**74.63**	**1.56**	**88.04****	**441**
Farm 3	Control	Arbor Acres	Male	15,912	42	-	3,230	76.90	1.57	95.68	469
	BA 5940	**Arbor Acres**	**Male**	**15,912**	**42**	**19-42**	**3,290**	**78.33**	**1.55**	**93.58**	**473**

BA 5940 treated groups have shown the better ***BW, ADG, FCR and PI*** compared to the Control groups, and economically outstanding of ROI.

Note

- *Dosages in BA 5940 programs were calculated based on 50 g/1,000 L. water and provided for 24 hours.*
- *Medication & vaccination of all groups in 3 farms followed farm's programs.*

***Remark ; Chronic Respiratory Disease outbreak in BA 5940 treated group*

Microbiome Analysis





16S rRNA Sequencing

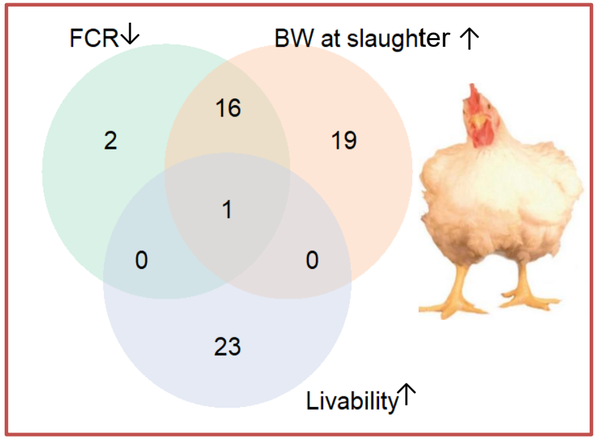
Microbiome composition is linked to performance KPIs

Bacterial biomarkers associated with feed efficiency, body weight, and livability

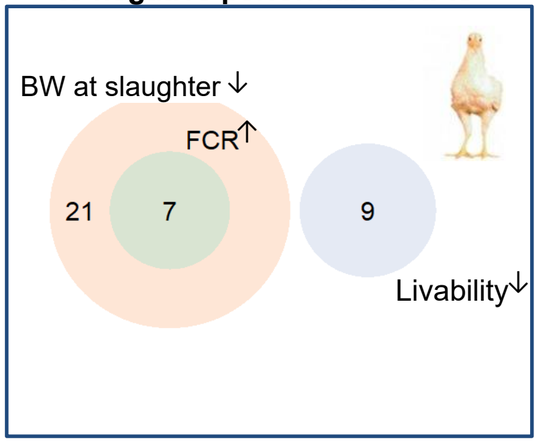
Why this matters

- Confirms microbiome modulation as a performance lever
- Creates biological rationale for *Bacillus amyloliquefaciens* CECT 5940 mode of action
- Focus on FCR, BW, livability = direct economic relevance

61 caecal biomarkers linked to positive performance KPIs



37 caecal biomarkers linked to negative performance KPIs



Core findings

- Body weight at slaughter significantly correlated with feed efficiency ($Q < 0.05$, corr = -0.66) → *expected*
- Livability negatively correlated with body weight at slaughter ($Q < 0.05$, corr = -0.33) → *expected*
- Multiple robust bacterial biomarkers linked to performance (multiple Spearman rank correlations, $Q < 0.05$)

Interpretation

- Performance is biologically programmable via microbiome targeting

BA 5940 (indirectly) improves livability via targeted microbiome modulation

Treatment induces specific microbiome shifts linked to improved livability

Why this matters

- Targeted modulation → predictable biological effect

Core findings

- **19** key bacterial biomarkers significantly modulated
- No disruption of the overall microbiome balance
- Very specific modulation of Bacteroidetes and Firmicutes without changing their overall abundance
- Most BA 5940 biomarkers are related to livability

Interpretation

- BA 5940 acts as a precision microbiome modulator,
  Not a broad disruptor

Tax-ID	Species	Ecobiol shifts abundance	FCR	BW at slaughter	Livability
ASV_01050	*Bacteroides uniformis*	up			increases
ASV_02737	*Coprobacter unclassified*	up	increases		decreases
ASV_01436	*Odoribacter splanchnicus*	up			increases
ASV_00093	*Parabacteroides distasonis*	up	decreases	increases	
ASV_00141	*Enterococcus cecorum*	up			
ASV_00027	*Parasutterella excrementihominis*	up			increases
ASV_00284	*Alistipes unclassified*	down	increases	increases	decreases
ASV_00179	*Bacteroides vulgatus*	down		decreases	decreases
ASV_00270	*Barnesiella intestinihominis*	down	increases	increases	decreases
ASV_00365	*Barnesiella viscericola*	down		decreases	
ASV_00029	*Parabacteroides johnsonii*	down	decreases	increases	decreases
ASV_00280	*Parabacteroides unclassified*	down		increases	
ASV_01025	*Parabacteroides unclassified*	down		increases	
ASV_07245	*Rikenella unclassified*	down			decreases
ASV_02042	*Helicobacter pullorum*	down		increases	
ASV_00049	*GCA-900066575 unclassified*	down		increases	
ASV_00260	*Lactobacillus unclassified*	down		decreases	
ASV_02200	*Megamonas unclassified*	down		increases	
ASV_00275	*Sutterella unclassified*	down		increases	decreases

Significant after FDR correction Q < 0.05

Significant before FDR correction P < 0.05

= no correlation

results of correlation analyses (ASVs and performance variables)

Gut flora stabilizer:
Bacillus amyloliquefaciens CECT 5940 (BA 5940)

Benefits

Can improve the FCR in broiler chicken

Can replace AGPs

Growth & Persistency

Spores germinate consistently under intestinal conditions in chicken and persist in the gut for minimum 3 days

Modes of action

Production of secondary metabolites & lactic acid

Excretion of digestive enzymes

Blocking of pathogenic bacterial communication (quorum quenching)

Modulation of the immune system

Inhibition

Can inhibit growth of relevant pathogens

Clostridium perfringens, Salmonella spp., E.coli

Gut Microbial Modulation

BA 5940 selectively modulates microbiome taxa associated with better desirable flock performance

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